

Anti-Inpainting: A Proactive Defense Approach Against Malicious Diffusion-Based Inpainters Under Unknown Conditions

Yimao Guo , Zuomin Qu , Wei Lu , *Member, IEEE*, and Xiangyang Luo , *Member, IEEE*

Abstract—The transferability of adversarial examples remains a critical challenge in the field of proactive defense. However, in the specific context of proactive defense against image inpainting, most existing work primarily focuses on the transferability of adversarial examples across different models, overlooking their transferability across diverse masks. To address this, we propose Anti-Inpainting, a proactive defense approach that achieves protection comprising three novel modules. First, we introduce a multi-level deep feature extractor to obtain intricate features from the diffusion denoising process, enhancing protective effectiveness. Second, we design a multi-scale, semantic-preserving data augmentation technique to enhance the transferability of adversarial perturbations across unknown conditions. Finally, we propose a selection-based distribution deviation optimization strategy to bolster protection against manipulations guided by diverse random seeds. Extensive experiments on InpaintGuardBench and CelebA-HQ demonstrate that Anti-Inpainting effectively defends against diffusion-based inpainters under unknown conditions. Additionally, our approach demonstrates robustness against various image purification methods and transferability across different diffusion model versions.

Index Terms—Proactive defense, adversarial attacks, diffusion model, image inpainting.

I. INTRODUCTION

IN RECENT years, deep generative models, particularly Denoising Diffusion Probabilistic Models (DDPMs) and Latent Diffusion Models (LDMs), have achieved breakthrough progress in the field of artificial intelligence [1], [2]. Leveraging their unique mechanism of iterative denoising within a compressed latent space, these models exhibit exceptional capabilities in high-fidelity image synthesis and fine-grained control. Particularly in the task of image inpainting, diffusion models allow users to precisely specify editing regions by providing masks, thereby generating high-quality manipulated results that seamlessly blend with the surrounding visual context [3], [4].

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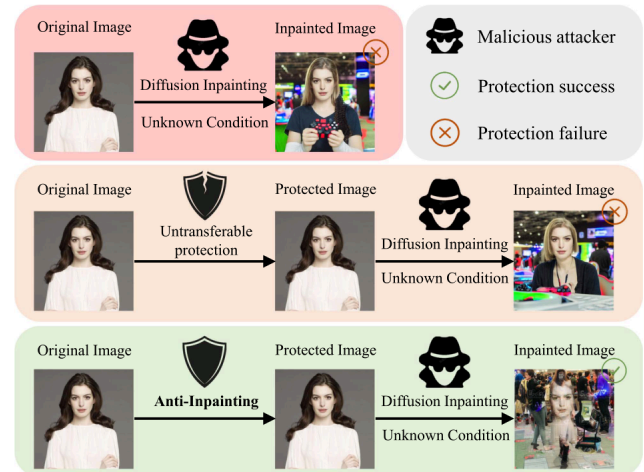


Fig. 1. The proactive defense against the misuse of diffusion models guided by unknown conditions.

However, the widespread adoption of these powerful image manipulation technologies has simultaneously raised severe ethical and security concerns [5], [6], [7], [8], as shown in Fig. 1 (top row). Given the unprecedented ability of diffusion models to generate highly deceptive and hyper-realistic visual content, malicious attackers can easily exploit open-source models (e.g., Stable Diffusion [9]) to fabricate news scenarios, manufacture fake evidence, or generate misleading imagery. As the editing capabilities of these models continue to evolve, developing robust safeguards to thwart such malicious misuse has become an urgent and critical issue in the realm of multimedia security [10], [11], [12], [13].

To counter this security threat, proactive defense has emerged as a highly promising protection strategy. The core principle of these methods involves injecting imperceptible adversarial perturbations into original images before they are published or shared. When malicious users attempt to edit these protected images using diffusion models, these stealthy perturbations disrupt the model's denoising generation process, leading to conspicuous visual artifacts or semantic collapse in the generated images, thereby preventing unauthorized editing.

Although existing proactive defense methods (e.g., PhotoGuard [14]) have demonstrated favorable protective effects in specifically controlled environments, they often falter when

confronted with the complex and unpredictable "unknown conditions" inherent in real-world scenarios, as shown in Fig. 1 (middle row). Specifically, current research suffers from two critical limitations. First, existing methods largely overlook the transferability of adversarial perturbations across diverse masks. Attackers in real-world settings can sketch unknown masks of arbitrary shapes based on their specific needs. Existing defensive perturbations are often heavily overfitted to the known masks used during training; thus, their protective efficacy significantly degrades when encountering manually created, unseen masks. While some works attempt to enhance robustness by shrinking mask contours, this approach often disrupts the semantic integrity of the masks, thereby weakening the overall protection. Second, existing works fail to adequately address the vulnerabilities introduced by the randomness of the initial latent state. During the editing process of diffusion models, the initial latent state serves as a crucial random variable input to the U-Net, exerting a decisive influence on the final noise prediction. Current defense methods typically optimize perturbations only under a fixed or limited set of latent states. This allows malicious attackers to easily bypass existing defense mechanisms and generate high-quality tampered images simply by resampling the random seed.

To overcome these limitations and achieve robust defense under unknown conditions, we argue that defensive strategies should not be confined solely to the terminal noise prediction output. Instead, they must delve into the internal, multi-level feature vulnerabilities of the diffusion process. Furthermore, it is imperative to augment mask distributions without compromising semantics, and to introduce distribution optimization strategies to suppress defense failures caused by latent state randomness. Based on these insights, this paper proposes a novel proactive defense approach named Anti-Inpainting, designed to provide robust protection for images against malicious diffusion-based inpainting manipulations under unknown conditions, as depicted in Fig. 1 (bottom row).

Specifically, our approach comprises three collaborative, innovative modules. First, unlike traditional attacks that focus merely on the final predicted noise, we introduce a multi-level deep feature extractor. This module directly targets crucial hierarchical features during the U-Net's downsampling and upsampling processes, fundamentally disrupting the model's inherent denoising capability. Second, to counter unseen malicious masks, we design a multi-scale, semantic-preserving data augmentation technique. This technique ensures semantic consistency while expanding mask diversity, thereby significantly enhancing the cross-condition transferability of adversarial perturbations. Finally, to address the challenge of defense failures caused by random latent states, we propose a selection-based distribution deviation optimization strategy. This strategy intelligently identifies and filters out specific latent states prone to causing protection failures, concentrating the limited perturbation budget on these "high-risk" states for targeted optimization. This effectively mitigates the impact of generation randomness.

We summarize our main contributions as follows:

- We propose Anti-Inpainting, a proactive defense approach that generates adversarial perturbations to protect images

against diffusion-based inpainting models under unknown conditions.

- We introduce a multi-level feature extractor to capture hierarchical image features. To enhance the transferability of perturbations, we design a multi-scale, semantic-preserving data augmentation. Furthermore, we develop a selection-based distribution deviation optimization strategy to ensure both effective protection and efficient optimization.
- Extensive experiments demonstrate that our proposed Anti-Inpainting effectively safeguards images against various diffusion-based inpainting models and exhibits strong robustness to diverse image purification techniques.

II. RELATED WORK

A. Adversarial Transferability

Adversarial examples, meticulously crafted inputs designed to deceive machine learning models, have exposed significant vulnerabilities in deep neural networks [15], [16], [17], [18], [19], [20], [21]. While transparent attacks effectively mislead models by fully accessing their gradients and architectures, real-world deployment typically operates under an opaque setting, where the attacker has zero knowledge of the target model's internal parameters or operating conditions. Consequently, improving the transferability of adversarial examples—the ability of an attack generated on a surrogate model to successfully deceive unseen target models or operate under unknown conditions—has become a paramount objective in adversarial attack. The fundamental challenge of transferability lies in the tendency of adversarial perturbations to overfit the specific decision boundaries or input constraints (e.g., predefined masks or viewpoints) of the surrogate environment.

To mitigate this overfitting and boost transferability, researchers have extensively explored input transformation and data augmentation techniques during the adversarial optimization process. Methods such as input diversity, translation-invariant attacks, and scale-invariant attacks apply random transformations to the input images at each iteration, forcing the perturbations to remain effective across various spatial states. However, when these traditional augmentation principles are directly applied to the domain of diffusion-based inpainting, they often encounter a semantic bottleneck. For instance, directly applying random distortions or severe contour-shrinking to the manipulation masks can fundamentally alter the topological structure and semantic meaning of the targeted region. This semantic degradation causes the optimized perturbations to misalign with the attacker's actual intent, ultimately reducing the defensive efficacy against unseen, manually crafted masks.

In contrast to conventional augmentation approaches, our work recognizes that preserving semantic integrity is the prerequisite for effective cross-condition transferability in proactive defense. We introduce a multi-scale, semantic-preserving data augmentation technique. By applying controlled morphological operations (dilation and erosion) to the masks during the generation of adversarial examples, our approach significantly diversifies the mask distribution without destroying the primary

semantic shape. This design explicitly prevents the defense from overfitting to a single known mask, thereby boosting the transferability of the protection against arbitrary, unknown masks deployed by malicious users.

B. Diffusion-Based Image Inpainting

Diffusion models have revolutionized high-fidelity image synthesis by conceptualizing generation as a gradual, iterative denoising process modeled as a Markov chain [22]. Latent Diffusion Models (LDMs) further advanced this paradigm by executing the diffusion process within a highly compressed latent space rather than the pixel space, significantly improving computational efficiency and generative quality [23], [24], [25]. The core engine driving this iterative denoising is the U-Net architecture, which predicts the noise added at each timestep [26], [27], [28], [29], [30]. During this process, the U-Net extracts hierarchical representations of the input, relying heavily on multi-level feature maps—from low-level structural details in the downsampling blocks to high-level semantic concepts in the upsampling blocks—to accurately reconstruct the image [31], [32], [33].

In the specific application of image inpainting, diffusion models offer fine-grained, user-directed manipulation [34], [35]. The generation process takes a complex joint input comprising the latent state, the masked image encoding, and the conditional manipulation mask. A critical yet often overlooked aspect of this process is the initial latent state, which is sampled from a standard normal distribution and serves as the starting point for the reverse denoising trajectory. Because the generative process is highly sensitive to this initial random seed, identical text prompts and masks can yield drastically different inpainted outcomes depending on the sampled latent state. Unfortunately, existing proactive defense mechanisms typically formulate their adversarial objectives based on the final predicted noise output under a fixed or strictly limited set of initial latent states. This myopic optimization strategy ignores the vulnerabilities hidden within the U-Net's intermediate feature hierarchy and leaves the defense highly susceptible to failure when an attacker simply resamples the initial random seed.

Diverging from existing terminal-noise attack paradigms, Anti-Inpainting fundamentally exploits the internal mechanics of diffusion-based inpainters. Recognizing that internal feature maps offer a more comprehensive attack surface than the final noise output, we introduce a multi-level deep feature extractor that targets critical block-wise features across the U-Net's encoder and decoder architectures. Furthermore, to neutralize the vulnerability associated with random seed resampling, we formulate a selection-based distribution deviation optimization strategy. This strategy dynamically evaluates the deviation of features across diverse latent states, concentrating the adversarial perturbation budget exclusively on viable candidates. By doing so, our approach not only disrupts the hierarchical semantic comprehension of the U-Net but also ensures robust protective performance against the inherent randomness of the diffusion process.

C. Proactive Defense Against Generative Models

Recently, proactive defense has emerged as a crucial mechanism to safeguard facial privacy and intellectual property against advanced generative model [36]. The core paradigm involves injecting imperceptible adversarial perturbations into the original images before they are distributed, thereby disrupting the generative process when malicious manipulation is attempted. Pioneering methods, such as PhotoGuard [14] and AdvDM [27], have demonstrated the feasibility of this approach by executing dual attacks on both the condition encoder and the diffusion generation stages [37], [38], [39]. These techniques typically optimize the perturbations to maximize the deviation in the latent space, aiming to induce significant visual artifacts or semantic collapse in the final synthesized images.

Despite their initial success, the efficacy of most existing proactive defenses heavily relies on prior knowledge of the attacker's operational conditions, severely limiting their real-world applicability. Current methods predominantly optimize adversarial perturbations based on predefined target masks and fixed initial latent states. However, in practical attack scenarios, malicious users can employ arbitrary, manually drawn masks to specify manipulation regions and easily resample the random noise seed (i.e., the initial latent state) to generate diverse outputs. Consequently, perturbations generated by existing methods tend to overfit the known training conditions. For instance, while DiffusionGuard [40] attempts to mitigate mask sensitivity via contour-shrinking mask augmentation, this heuristic operation inevitably compromises the semantic integrity of the mask, leading to suboptimal protection capabilities. Furthermore, the critical vulnerability introduced by the randomness of the initial latent state remains largely unaddressed in our work, leaving a significant backdoor for attackers to bypass the defense via simple resampling.

Unlike the aforementioned works, our approach systematically confronts the challenge of unpredictable unknown conditions in proactive defense. Rather than merely attacking the final noise prediction under fixed settings, we propose Anti-Inpainting, which strategically targets the multi-level deep features of the U-Net to fundamentally paralyze the model's inherent denoising capability. Furthermore, to break the bottleneck of condition overfitting, we introduce a multi-scale semantic-preserving data augmentation technique to ensure mask robustness without sacrificing semantic fidelity. Crucially, we propose a selection-based distribution deviation optimization strategy that intelligently mitigates the protective failures caused by latent state resampling. This comprehensive design ensures that the adversarial examples generated by our approach possess significantly stronger transferability and robustness in complex, real-world manipulation scenarios.

III. METHOD

A. Threat Model

Image inpainting, which involves modifying targeted regions within an image, is another significant application of generative

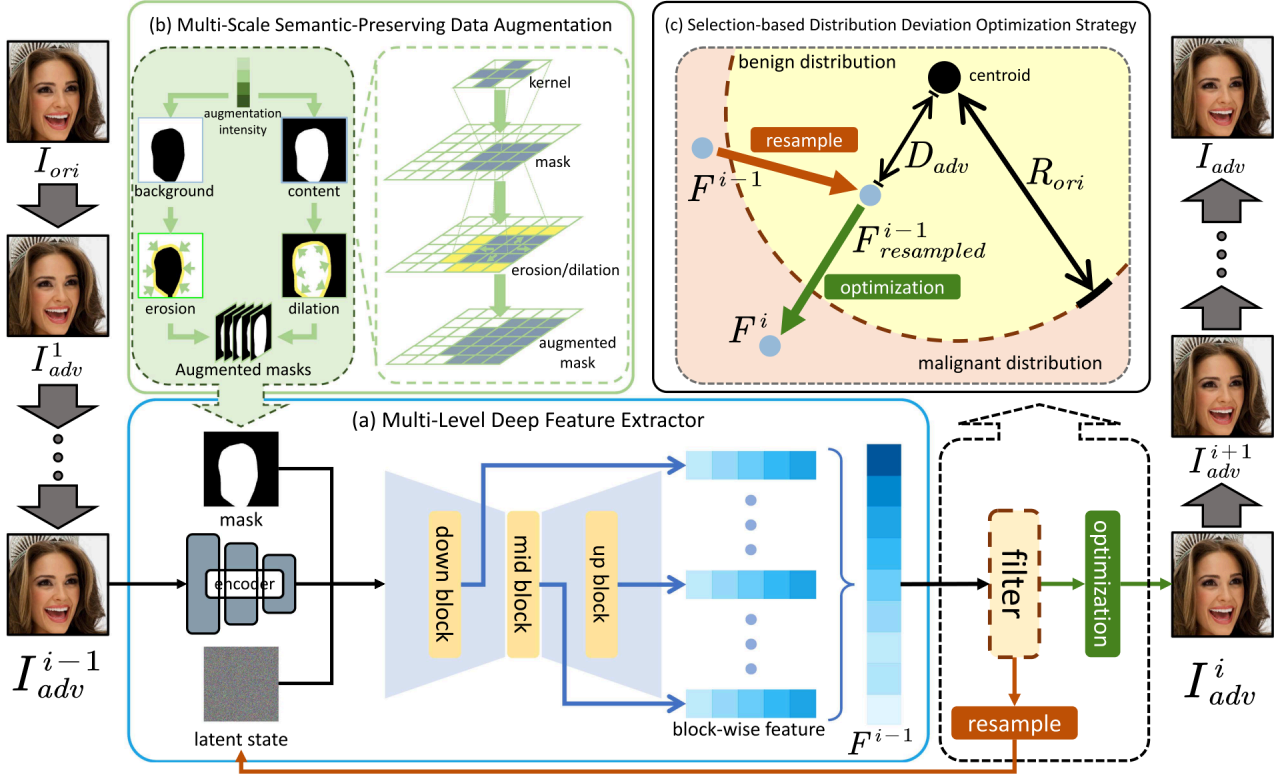


Fig. 2. Overview of Anti-Inpainting. We propose an iterative method to generate adversarial images from original images. In each iteration, diverse masks are first generated via multi-scale semantic-preserving data augmentation. These masks, along with the latent state and the adversarial image, are fed into a multi-level deep feature extractor. A selection-based distribution deviation optimization strategy then selects salient features from the extractor, which are subsequently used to update the adversarial image.

models. The process begins by applying a mask M to the manipulation region of a given image I . An image encoder $\mathcal{E}(\cdot)$ is then used to extract embeddings from I . In the subsequent diffusion process, these embeddings and the mask are concatenated with the latent state z_t , serving as input to the noise predictor $\epsilon_\theta(\cdot)$. This iterative denoising process for inpainting can be formulated as follows:

$$\begin{aligned} z_{t-1} &= \frac{1}{\sqrt{\alpha_t}} \left(z_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} n_{pred} \right) + \sigma_t n, \\ n_{pred} &= \epsilon_\theta(z_t, \mathcal{E}(I), M, t, \text{clip}(\mathcal{T})). \end{aligned} \quad (1)$$

where t denotes the timestep, α_t and $\bar{\alpha}_t$ are pre-defined hyperparameters, and $\text{clip}(\mathcal{T})$ represents the text embeddings for the manipulation prompt $\mathcal{T}$.

B. Task Formulation

The goal of proactive defense is to protect image privacy by disrupting unauthorized manipulations performed by diffusion models. Given a clean image I and a diffusion model, the adversarial perturbation δ is added into the clean image I . To ensure visual imperceptibility, the perturbation δ is typically limited using the norm bound η . To disrupt the reverse diffusion

process, the perturbation δ is optimized by:

$$\delta = \arg \max_{\|\delta\|_\infty \leq \eta} \|n_{pred} - \epsilon_\theta(z, \mathcal{E}(I + \delta), M, t, \text{clip}(\mathcal{T}))\|_2. \quad (2)$$

C. Overall Framework

To implement a robust proactive defense against malicious diffusion-based inpainting under unknown conditions, we propose the Anti-Inpainting framework, as illustrated in Fig. 2. Our approach builds upon the Projected Gradient Descent (PGD) framework [41] to craft imperceptible adversarial perturbations iteratively. Specifically, to overcome the limitations of existing methods, the proposed framework consists of three collaborative modules: a multi-scale semantic-preserving data augmentation, a multi-level deep feature extractor, and a selection-based distribution deviation optimization strategy.

First of all, the multi-scale semantic-preserving data augmentation is introduced to overcome the limitation of condition overfitting. During each iteration, this module dynamically generates diverse manipulation masks to expand the mask distribution without compromising their core semantic integrity. This step ensures that the crafted adversarial perturbations generalize effectively to unseen, manually drawn masks. Please refer to Section III-E for detailed information.

Second, the multi-level deep feature extractor plays a dominant role in fundamentally paralyzing the U-Net's inherent denoising capability. Instead of merely attacking the final noise prediction, we extract and target intermediate feature maps across the encoder and decoder blocks as the model processes the augmented data. By doing so, we expose a more comprehensive attack surface to disrupt the hierarchical semantic comprehension of the diffusion model. Please take a look at Section III-D for more technical details.

Finally, the selection-based distribution deviation optimization strategy is utilized to neutralize the vulnerability associated with random seed resampling. This strategy intelligently filters out high-risk latent states that typically lead to protection failures, allowing us to concentrate the perturbation budget on viable feature candidates. It computes the specific loss function to effectively update the adversarial image, thereby mitigating the negative impact of generative randomness. Please refer to Section III-F for more technical information.

D. Multi-Level Deep Feature Extractor

Mainstream adversarial attacks on diffusion models primarily target the final output of the U-Net, the predicted noise. This approach implicitly assumes the final prediction is a sufficient proxy for all crucial internal computations. We contend that the internal feature maps of the U-Net's encoder and decoder blocks offer a more comprehensive target. These maps represent a spectrum of features, from low-level patterns to high-level semantics, which are vital for the denoising process. By only attacking the final output, existing methods fail to fully exploit vulnerabilities within the model's feature hierarchy. Therefore, to achieve a more potent attack, we propose a multi-level deep feature extractor that captures block-wise features across the U-Net architecture, as shown in Fig. 2(a).

First, we construct the inputs for the first block of the U-Net module, denoted as ϵ^0 . At each denoising timestep t , the model takes two primary inputs: a main input tensor and a conditioning vector c . The main input tensor is formed by concatenating the initial latent state z , the VAE-encoded input image $\mathcal{E}(I_{input})$, and the input mask M_{input} . The conditioning vector c combines the timestep embedding for t and the text prompt embedding, $\text{clip}(\mathcal{T})$:

$$c = \text{concat}(t, \text{clip}(\mathcal{T})), \quad (3)$$

$$f_0 = \epsilon^0(z, \mathcal{E}(I_{input}), M_{input}, c), \quad (4)$$

where the latent state z is sampled from normal distribution, and timestep t is sampled from uniform distribution. We obtain the intermediate variable from the U-Net module, as follows:

$$f_i = \begin{cases} \epsilon_{down}^i(f_{i-1}, c), & \text{if } 5 > i > 0, \\ \epsilon_{mid}^i(f_{i-1}, c), & \text{if } i = 5, \\ \epsilon_{up}^i(f_{i-1}, f_{10-i}, c), & \text{if } 10 > i > 5, \end{cases} \quad (5)$$

where i is the number corresponding to the block ϵ . ϵ_{down}^i , ϵ_{mid}^i and ϵ_{up}^i denote the downsampling block, middle layer, and upsampling block of U-Net, respectively. And then f_{10} is the output of post-processing module ϵ^{10} . In addition, we combine the intermediate variable of each U-Net block and define the

whole process as multi-level deep feature extraction ϕ , which is defined as:

$$\begin{aligned} F &= \phi(z, \mathcal{E}(I_{input}), M_{input}, t, \text{clip}(\mathcal{T})), \\ &= \text{concat}(f_0, f_1, f_2, f_3, \dots, f_{10}). \end{aligned} \quad (6)$$

We compute the mean of the multi-level deep features, $\bar{F}_{ori}$, across multiple latent states z to serve as the feature representation for the original image I_{ori} . Additionally, we compute the distribution radius R_{ori} to quantify the dispersion of these features for the image I_{ori} :

$$\bar{F}_{ori} = E_{z \in N(0,1)} [\phi(z, \mathcal{E}(I_{ori}), M, t, \mathcal{T})], \quad (7)$$

$$R_{ori} = E_{z \in N(0,1)} [\|\phi(z, \mathcal{E}(I_{ori}), M, t, \mathcal{T}) - \bar{F}_{ori}\|_2]. \quad (8)$$

E. Multi-Scale Semantic-Preserving Data Augmentation

A key limitation of current methods is their reliance on known guidance conditions, leaving images vulnerable to the unpredictable and multifaceted manipulations employed by malicious users. While DiffusionGuard [40] enhances robustness against unknown conditions by using augmented masks, its contour-shrinking technique compromises the mask's semantic integrity, thus weakening the overall protection. To overcome this, we introduce multi-scale semantic-preserving data augmentation, as shown in Fig. 2(b). Our approach enhances the diversity of masks used in adversarial optimization without sacrificing their semantic information:

$$M_{aug} = \begin{cases} \Omega(M, n) & \text{if } n \geq 0, \\ \zeta(M, -n) & \text{if } n < 0, \end{cases} \quad (9)$$

where $\Omega(\cdot)$ is mask dilation operation, and $\zeta(\cdot)$ is mask erosion operation. We introduce a data augmentation scheme for the input mask M . The process is governed by an integer n sampled uniformly from $[-\gamma, \gamma]$, where γ is the augmentation intensity hyperparameter. The augmented mask M_{aug} is obtained by applying either a morphological dilation (for $n > 0$) or erosion (for $n < 0$) to M using a square kernel of size $|n| \times |n|$. By applying the moderate dilation or erosion, we maintain the mask's overall topology and primary shape, ensuring it continues to represent the same semantic object while introducing boundary variations for adversarial optimization.

F. Selection-Based Distribution Deviation Optimization Strategy

To maximize the protective effect under a fixed perturbation budget η , we address the issue of inefficient budget allocation. We posit that optimizing against adversarial features that have already deviated drastically from the original distribution yields diminishing returns. Therefore, we introduce a selection mechanism to focus the optimization on more impactful features. Specifically, instead of indiscriminately optimizing against all adversarial features, we focus the optimization process only on the adversarial features that remain within a defined proximity of the benign feature distribution. By avoiding budget allocation to features that have already deviated significantly, we can achieve a more potent and robust protective effect.

To implement this strategy, we first characterize the benign feature space. As illustrated in Fig. 2(c), we take the original clean image and generate a set of benign variants. These variants are then passed through the deep feature extractor ϕ to obtain a collection of benign features. From this collection, we compute the centroid $\bar{F}_{ori}$ and a boundary threshold R_{ori} , which together define the boundary of our target benign distribution. In each optimization step i , this benign distribution is used as a reference. We take the adversarial image from the previous iteration, I_{adv}^{i-1} , and extract its corresponding adversarial feature using the same extractor ϕ :

$$F^{i-1} = \phi(z, \mathcal{E}(I_{adv}^{i-1}), M_{aug}, t, \mathcal{T}), \quad (10)$$

where z is the latent state. Our selection mechanism operates as a conditional filter within each optimization iteration i . For each adversarial feature sample, generated using a latent state z , we first determine its viability for optimization. This is done by comparing its distance from the benign centroid, $D_{adv} = \|F^i - \bar{F}_{ori}\|_2$, against a dynamic threshold, $\tau \cdot R_{ori}$, where τ is a hyperparameter. If the feature is outside the boundary ($D_{adv} > \tau \cdot R_{ori}$), as exemplified by the red point in Fig. 2, we consider it an inefficient candidate for optimization. We discard this sample and resample a new latent state, z , to generate a new feature. This process repeats until a viable candidate is found or a maximum number of resampling attempts is reached. This prevents wasting the perturbation budget on features that have already diverged excessively. Conversely, if a feature lies within the boundary, it is deemed eligible for the adversarial attack. The goal of the attack is to push this feature away from the benign distribution. To achieve this, we define our loss function to maximize the distance between the adversarial feature F^i and the benign centroid $\bar{F}_{ori}$, as shown below:

$$L_{adv} = -\|(F^i - \bar{F}_{ori})\|_2^2. \quad (11)$$

This loss guides the update of the adversarial image. In each step, the gradient of L_{adv} with respect to the adversarial image is computed and used to perform the update. The complete pseudocode for our Anti-Inpainting adversarial optimization is presented in Algorithm 1.

IV. EXPERIMENTS

A. Experimental Setup

a) Datasets: We conduct quantitative evaluations of our approach and competing methods on the InpaintGuardBench [40] and CelebA-HQ [42] datasets. InpaintGuardBench consists of 42 images, each containing one known mask, four unknown masks, and 10 text prompts. For CelebA-HQ, we select the first 100 images for testing. For each of these images, we use the corresponding skin mask from CelebAMask-HQ [43] as the known mask and manually generated four unknown masks. These masks are created manually by either drawing with a circular brush or overlaying simple geometric shapes. Finally, all masks used in the quantitative experiments will be made publicly available on our project repository.

b) Comparison Methods: We compare six adversarial proactive defense methods for diffusion models:

Algorithm 1: Anti-Inpainting Adversarial Optimization.

Require: Clean image I_{ori} , mask M , prompt embedding $\text{clip}(\mathcal{T})$, budget η , step size α , iterations K , augmentation bound b , threshold τ , resample limit S .

Ensure: Adversarial image I_{adv} .

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1:  $I_{adv}^0 \leftarrow I_{ori}$ 
2: // Stage 1: Compute benign feature distribution
3: Sample multiple latent states  $\{z_j\}_{j=1}^J \sim \mathcal{N}(0, \mathbf{I})$ 
4: Extract deep features
    $F_j \leftarrow \phi(z_j, \mathcal{E}(I_{input}), M_{input}, t, \text{clip}(\mathcal{T}))$  for all  $j$ 
5: Compute centroid  $\bar{F}_{ori}$  and distribution radius  $R_{ori}$  via
   (7) and (8)
6: // Stage 2: Iterative Adversarial Optimization
7: for  $i = 1$  to  $K$  do
8:   Module 1: Semantic-Preserving Data Augmentation
9:   Sample intensity  $n \sim \mathcal{U}[-b, b]$ 
10:  Generate  $M_{aug} \leftarrow \Omega(M, n)$  or  $\zeta(M, -n)$  via (9)
11:   $valid\_candidate \leftarrow \text{False}$ ,  $s \leftarrow 0$ 
12:  Module 2 & 3: Feature Extraction and Selection
13:  while not  $valid\_candidate$  and  $s < S$  do
14:    Sample latent state  $z \sim \mathcal{N}(0, \mathbf{I})$ 
15:     $F^i \leftarrow \phi(z, \mathcal{E}(I_{adv}^{i-1}), M_{aug}, t, \text{clip}(\mathcal{T}))$  via (10)
16:     $D_{adv} \leftarrow \|F^i - \bar{F}_{ori}\|_2$ 
17:    if  $D_{adv} \leq \tau \cdot R_{ori}$  then
18:       $valid\_candidate \leftarrow \text{True}$ 
19:    else
20:       $s \leftarrow s + 1$   $\triangleright$  Resample to avoid inefficient
        optimization
21:    end if
22:  end while
23:  // Update adversarial perturbation
24:   $L_{adv} \leftarrow -\|F^i - \bar{F}_{ori}\|_2^2$  via (11)
25:   $I_{adv}^i \leftarrow I_{adv}^{i-1} - \alpha \cdot \text{sign}(\nabla_{I_{adv}} L_{adv})$ 
26:   $I_{adv}^i \leftarrow \text{clip}_{[I_{ori}-\eta, I_{ori}+\eta]}(I_{adv}^i)$ 
27: end for
28: return  $I_{adv}^K$ 

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- *PhotoGuard* [14]: disrupt an image's latent representations from VAE encoder, protecting it from unauthorized malicious AI editing.
- *AdvDM* [27]: Optimizes adversarial noise by maximizing the prediction loss during the reverse denoising process, misleading the diffusion model into generating meaningless or distorted outputs.
- *MFA* [27]: Exploits the model's high sensitivity to the mean of input noise by introducing mean fluctuations during the reverse generation process to corrupt the final visual output.
- *Mist*: Combines textual and semantic adversarial losses to create a perturbation sample.
- *DiffusionGuard*: Specifically targets mask-based inpainting tampering by concentrating adversarial interference on the early, critical denoising stages of the diffusion process to disable malicious image restoration.
- *DDD* [44]: Efficiently disrupts inpainting synthesis at a low computational cost by targeting the most vulnerable

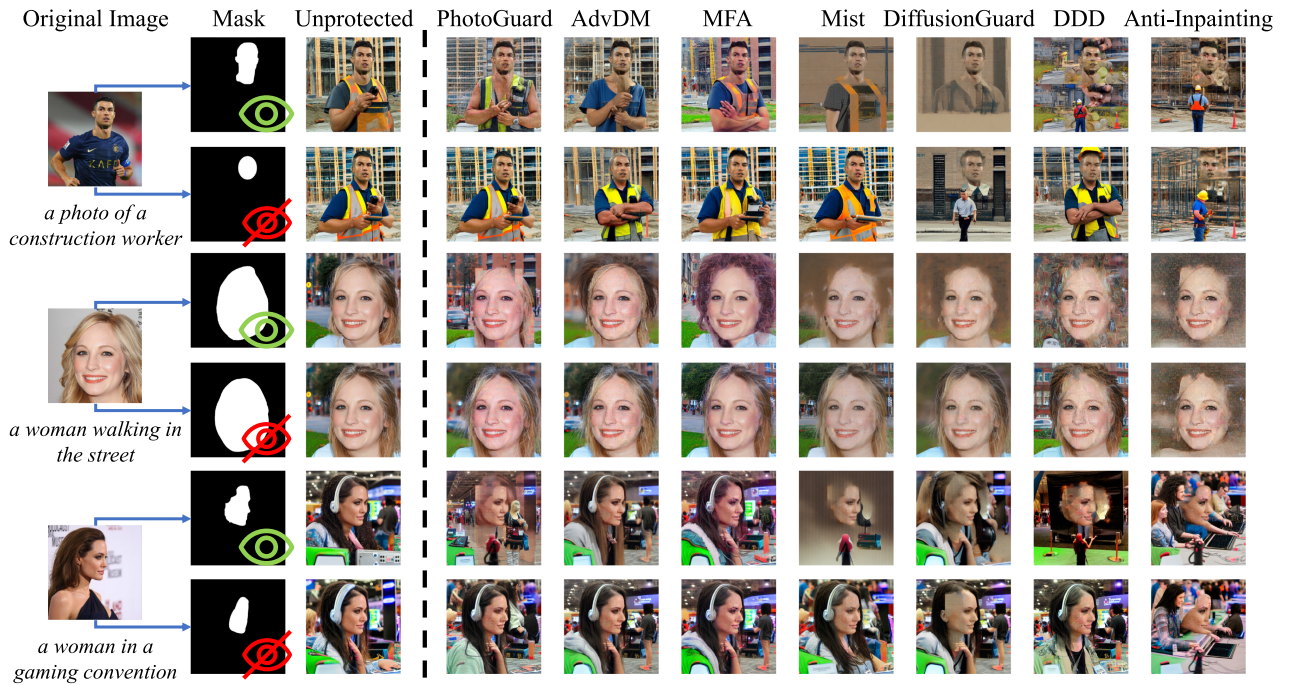


Fig. 3. The qualitative results between comparison methods and Anti-Inpainting. The text below each original image is the prompt used to generate the corresponding forged image. The green eye icon on the mask indicates that the mask is known during adversarial example generation, while the red, crossed-out eye icon signifies that it is unknown.

timesteps and guiding latent features toward a semantic digression away from the original meaning.

To simulate real-world scenarios, we generate the protected images using skin masks and empty text prompts. Subsequently, we evaluate the protective performance of these images against inpainting attacks guided by manual masks and malicious text prompts.

c) Evaluation Metrics: We assess the impact of adversarial perturbations on diffusion-based inpainting models using three sets of metrics. To quantify the difference between the protected and unprotected inpainting results under the same random seed, we compute PSNR, SSIM [45], and LPIPS [46]. The visual quality of the resulting images is evaluated via the ImageReward (IR) score [47]. Lastly, the ArcFace similarity (ARC) [48] is calculated to evaluate the preservation of facial identity information.

d) Implementation Details: The perturbation is constrained under the L-infinity norm with a magnitude of 16/255. For each sample, we perform 800 optimization iterations. Our primary attack target is the Runway v1.5 diffusion-based inpainter. To evaluate transferability, we also test the generated adversarial samples on the Stability AI v2.0 inpainter [9]. All experiments are conducted on NVIDIA 3090 GPUs, and our approach requires up to 16 GB of GPU memory.

B. Comparative Experiment

a) Qualitative Results: Fig. 3 presents the original images, their corresponding masks, and the resulting inpainted images. The figure also displays the inpainting results from adversarial

examples generated by both the compared methods and our proposed approach. As shown, the compared methods are effective under known conditions but fail under unknown ones. In contrast, our approach demonstrates strong protective performance in both scenarios. These qualitative results validate our conclusion that increasing mask diversity during adversarial training and strategically selecting the initial latent state improves the transferability of adversarial examples to unknown conditions.

b) Quantitative Results and Analysis: To comprehensively evaluate the effectiveness of the proposed Anti-Inpainting method, we analyze its performance against state-of-the-art proactive defense baselines under both standard and stringent malicious manipulation scenarios. The quantitative results on the InpaintGuardBench and CelebA-HQ datasets are summarized in Tables I and II. Throughout our analysis, a lower PSNR, SSIM, IR (ImageReward), and ARC (ArcFace similarity) signify a stronger disruption of the unauthorized editing process, whereas a higher LPIPS indicates greater perceptual distortion in the tampered image.

Table I presents the protective efficacy of adversarial examples when subjected to unknown mask conditions under a single initial latent state. The experimental results clearly demonstrate that Anti-Inpainting exhibits an exceptional level of transferability and protective strength. When compared to the established PhotoGuard, our approach significantly degrades the quality of the unauthorized generation. For instance, on the CelebA-HQ dataset, Anti-Inpainting achieves a PSNR of 14.704 and an SSIM of 0.468, representing a substantial drop of 17.7% and 26.8% respectively, compared to PhotoGuard (PSNR: 17.874, SSIM: 0.640).

TABLE I

THE QUANTITATIVE RESULTS OF COMPARISON METHODS AND ANTI-INPAINTING. THE BEST ATTACKING PERFORMANCES OF METHODS ARE MARKED AS BOLD. NOTE: LOWER VALUES FOR PSNR, SSIM, IR, AND ARC ($\downarrow$), AND HIGHER VALUES FOR LPIPS ($\uparrow$) INDICATE A STRONGER DEFENSE PERFORMANCE.

Methods	InpaintGuardBench					CelebA-HQ				
	PSNR $\downarrow$	SSIM $\downarrow$	LPIPS $\uparrow$	IR $\downarrow$	ARC $\downarrow$	PSNR $\downarrow$	SSIM $\downarrow$	LPIPS $\uparrow$	IR $\downarrow$	ARC $\downarrow$
PhotoGuard	16.518	0.600	0.404	-0.015	0.674	17.874	0.640	0.380	-0.011	0.861
AdvDM	16.695	0.598	0.402	-0.032	0.677	18.000	0.600	0.393	-0.011	0.836
MFA	16.975	0.609	0.392	-0.032	0.739	19.372	0.686	0.285	-0.010	0.923
Mist	15.687	0.533	0.481	-0.274	0.635	16.711	0.551	0.457	-0.132	0.809
DiffusionGuard	14.797	0.477	0.576	-0.578	0.571	15.874	0.495	0.588	-1.617	0.762
DDD	14.390	0.488	0.520	-0.224	0.548	15.779	0.525	0.491	-1.369	0.777
Anti-Inpainting	12.875	0.414	0.595	-0.473	0.491	14.704	0.468	0.592	-1.658	0.744

TABLE II

THE QUANTITATIVE RESULTS OF COMPARISON METHODS AND ANTI-INPAINTING UNDER MULTIPLE INITIAL LATENT STATES. NOTE: LOWER VALUES FOR PSNR, SSIM, IR, AND ARC ($\downarrow$), AND HIGHER VALUES FOR LPIPS ($\uparrow$) INDICATE A STRONGER DEFENSE PERFORMANCE.

Methods	InpaintGuardBench					CelebA-HQ				
	PSNR $\downarrow$	SSIM $\downarrow$	LPIPS $\uparrow$	IR $\downarrow$	ARC $\downarrow$	PSNR $\downarrow$	SSIM $\downarrow$	LPIPS $\uparrow$	IR $\downarrow$	ARC $\downarrow$
PhotoGuard	19.964	0.722	0.297	-0.661	0.877	20.580	0.761	0.243	0.975	0.692
AdvDM	19.751	0.687	0.305	-0.717	0.853	20.979	0.770	0.234	0.920	0.692
MFA	22.422	0.796	0.186	-0.640	0.938	21.097	0.772	0.235	0.933	0.749
Mist	18.527	0.668	0.339	-0.786	0.831	19.255	0.698	0.311	0.877	0.641
DiffusionGuard	17.965	0.625	0.434	-1.076	0.778	18.241	0.647	0.386	0.729	0.604
DDD	21.753	0.770	0.201	-0.606	0.936	17.509	0.652	0.378	0.881	0.567
Anti-Inpainting	17.665	0.618	0.408	-1.012	0.805	15.619	0.573	0.466	0.663	0.476

Furthermore, our approach yields the most profound degradation in facial identity preservation, achieving an ARC score of 0.744 on CelebA-HQ, significantly lower than the 0.861 achieved by PhotoGuard. This pronounced superiority indicates that while traditional methods strictly overfit to known mask boundaries, our multi-scale semantic-preserving data augmentation effectively generalizes the adversarial perturbations. By learning robust features across varied mask scales during optimization, our protected images maintain potent defensive capabilities even when attackers employ completely unseen, manually drawn masks.

c) Robustness against Latent State Resampling: In real-world scenarios, sophisticated attackers frequently attempt to bypass static defenses by iteratively resampling the initial latent state (random seed) until a satisfactory tampered image is generated. Table II rigorously evaluates this severe threat by reporting the most successful malicious outcomes across multiple random seeds.

Under these stringent conditions, the protective efficacy of most baseline methods experiences a precipitous collapse. For example, PhotoGuard’s PSNR on the InpaintGuardBench dataset surges from 16.518 (Table I) to 19.964 (Table II), indicating a severe failure to disrupt the generation. Similarly, DiffusionGuard struggles to maintain its defense, yielding an SSIM of 0.625.

In stark contrast, Anti-Inpainting exhibits remarkable stability and resilience. It consistently maintains the lowest PSNR (17.665) and SSIM (0.618) on InpaintGuardBench, alongside the lowest ImageReward (IR) score (-1.012), which vividly reflects the poor visual quality of the attacker’s final output. This

TABLE III

THE COMPUTATIONAL COST OF COMPARISON METHODS AND ANTI-INPAINTING

Method	runtime(s)	GPU memory(MB)
photoguard	341.43	13841
advDM	252.36	11241
MFA	276.31	11263
mist	239.29	11871
ddd	205.26	17049
ours	180.00	13275

robust performance firmly validates the necessity and effectiveness of our selection-based distribution deviation optimization strategy. By intelligently filtering and heavily penalizing those “high-risk” latent states that typically lead to protection breakdown, our approach ensures that the defensive perturbation remains universally potent despite the stochasticity of the diffusion denoising process.

Overall, the consistent margin by which Anti-Inpainting outperforms advanced techniques like Mist and DDD across both datasets underscores the holistic superiority of our attack strategy. By targeting hierarchical structural features via the multi-level deep feature extractor—rather than merely attacking the terminal noise prediction—our approach guarantees a fundamental, irreversible disruption of the diffusion model’s image comprehension, striking an optimal balance between defense transferability and robustness.

d) Computational Cost: We benchmarked the computational cost on InpaintGuardBench. As detailed in Table III, our approach, enabled by a selection-based distribution deviation optimization, records the fastest GPU execution time while

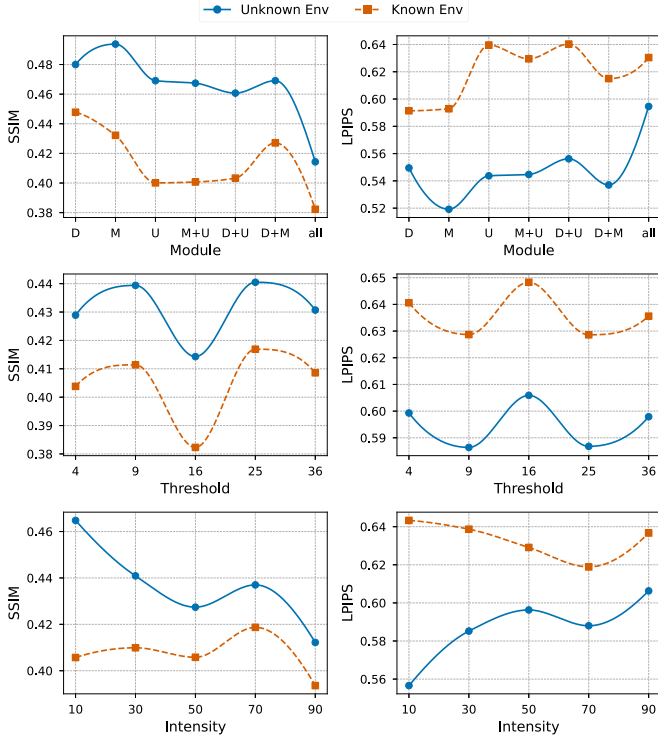


Fig. 4. The results of ablation experiments on feature selection, augmentation intensities, and optimization thresholds.

preserving a GPU memory footprint comparable to that of competing methods. Because we sum the scalar L2 distances rather than backpropagating through an artificially concatenated dense vector, the computational overhead is highly manageable and practical. The calculation of the L2 distance at each layer only involves element-wise subtraction and squaring, which is computationally negligible. During backpropagation, standard deep learning frameworks efficiently route the gradients from these intermediate loss terms back through the existing computational graph of the U-Net. Therefore, the overall memory consumption and time complexity are essentially equivalent to a standard forward and backward pass of the diffusion model's U-Net, without introducing heavy additional bottlenecks.

C. Ablation Study

a) Feature Selection: We conducted an ablation study to evaluate the effectiveness of using multi-level features from the diffusion model for adversarial protection. The U-Net was divided into three components: downsampling blocks $\mathcal{D}$, a middle block $\mathcal{M}$, and upsampling blocks $\mathcal{U}$. We then created several control groups by combining features from these respective components. The first row of Fig. 4 reveals the crucial role of features across all U-Net levels. Specifically, both low-level features from the downsampling path and high-level features from the upsampling path proved essential for generating adversarial examples. This finding underscores the importance of the multi-level feature extractor in our approach. While optimizing only the terminal noise prediction targets the final generation

TABLE IV
THE PROTECTIVE PERFORMANCE OF COMPARISON METHODS AND ANTI-INPAINTING AGAINST THE DIFFERENT VERSION OF DIFFUSION-BASED INPAINTERS ON INPAINTGUARDBENCH. THE EVALUATION IS PERFORMED UNDER THE SAME UNKNOWN MASK CONDITIONS (AND VARYING RANDOM SEEDS) AS DETAILED IN TABLE I. NOTE: LOWER VALUES FOR PSNR, SSIM, IR, AND ARC ($\downarrow$), AND HIGHER VALUES FOR LPIPS ($\uparrow$) INDICATE A STRONGER DEFENSE PERFORMANCE.

Methods	InpaintGuardBench				
	PSNR $\downarrow$	SSIM $\downarrow$	LPIPS $\uparrow$	IR $\downarrow$	ARC $\downarrow$
PhotoGuard	16.385	0.610	0.397	0.147	0.685
AdvDM	16.354	0.604	0.398	0.102	0.677
MFA	16.762	0.623	0.380	0.172	0.749
Mist	15.185	0.529	0.482	-0.020	0.659
DiffusionGuard	14.500	0.485	0.550	-0.251	0.594
DDD	14.305	0.506	0.503	0.010	0.567
Anti-Inpainting	13.364	0.458	0.565	-0.131	0.532

output, it often overfits to the specific mask or seed seen during optimization. In contrast, diffusion models process semantic information hierarchically. By extracting and disrupting features across multiple levels—down-sampling, middle, and up-sampling blocks—our approach corrupts the image's semantic structure at varying granularities. This deep, structural disruption makes the adversarial perturbations significantly harder for the model to bypass when hallucinating content under unknown and unseen conditions.

b) Optimization Thresholds: This ablation study investigates the impact of the optimization threshold on the efficacy of adversarial examples. We discovered a U-shaped performance curve: trivially small thresholds would result in futile optimization, while excessively large ones would overlook valuable initial latent states, both diminishing performance. Significantly, we discovered a strong correlation between the unseen mask and seen mask performance of the adversarial examples across various thresholds, as shown in the second row of Fig. 4. This correlation allows us to use the more readily available seen mask metrics as a proxy for tuning the optimization threshold, thereby maximizing the success rate of unseen mask attacks.

c) Augmentation Intensities: In this ablation study, we evaluated the effect of data augmentation intensity on adversarial example performance. The third row of Fig. 4 indicates that with increasing data augmentation intensity, the seen mask performance initially declines before rising. In contrast, the unseen mask performance exhibits a consistent upward trend. This trend in the seen mask setting suggests that our augmentation module does more than simply enhance unseen mask transferability; it fundamentally improves the adversarial examples' ability to interfere with the diffusion model's image comprehension.

D. Transferability Study

We assessed the transferability of adversarial examples generated by our proposed approach from Runway's v1.5 model to StableAI's v2.0 model. It is noteworthy that while these two models share an identical architecture, they are trained under different protocols. As shown in Table IV, our approach demonstrates state-of-the-art performance on the InpaintGuardBench

TABLE V
THE QUANTITATIVE RESULTS OF COMPARISON METHODS AND ANTI-INPAINTING UNDER JPEG COMPRESSION

Attack Method	iters	L_{∞} -eps	JPEG Compression (90)				JPEG Compression (70)				JPEG Compression (50)			
			White-box Mask		Black-box Mask		White-box Mask		Black-box Mask		White-box Mask		Black-box Mask	
			PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓
photoguard	800	16/255	15.8517	0.5798	17.7154	0.6555	17.6707	0.6512	18.7656	0.6967	18.4247	0.6875	19.4419	0.7271
advDM	800	16/255	16.5598	0.575	17.9891	0.6588	18.0735	0.6529	19.1124	0.7078	18.8836	0.6971	19.7478	0.7343
MFA	800	16/255	16.5322	0.5678	18.1513	0.663	18.472	0.6701	19.6726	0.7282	19.4428	0.7174	20.2125	0.7521
MIST	800	16/255	16.1482	0.5393	17.5153	0.6311	18.1537	0.651	19.1635	0.7073	19.4504	0.7161	20.1323	0.7496
diffusionguard	800	16/255	15.9571	0.5321	16.7161	0.5909	17.8	0.6334	18.4604	0.6775	19.1841	0.7064	19.8462	0.7385
ddd	800	16/255	15.1715	0.5153	16.4214	0.5911	16.8485	0.6064	17.9394	0.6629	18.1012	0.6686	18.9807	0.7086
Anti-Inpainting	800	16/255	14.1796	0.4749	14.6355	0.514	15.716	0.5577	16.2395	0.5896	17.0001	0.6203	17.3869	0.6448

TABLE VI
THE QUANTITATIVE RESULTS OF COMPARISON METHODS AND ANTI-INPAINTING UNDER BIT COMPRESSION

Attack Method	iters	L_{∞} -eps	Bit Compression (4-bit)				Bit Compression (3-bit)				Bit Compression (2-bit)				Bit Compression (1-bit)			
			White-box Mask		Black-box Mask		White-box Mask		Black-box Mask		White-box Mask		Black-box Mask		White-box Mask		Black-box Mask	
			PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓
photoguard	800	16/255	16.1888	0.5723	18.2433	0.6611	14.3067	0.51	17.0339	0.6208	13.9966	0.4994	16.6192	0.605	13.9196	0.4958	16.5217	0.6007
advDM	800	16/255	16.7237	0.5635	18.1895	0.6515	15.3779	0.503	17.2239	0.6202	15.0982	0.4871	16.8338	0.6039	14.9945	0.4828	16.7201	0.5987
MFA	800	16/255	16.266	0.5391	18.4186	0.6585	15.1133	0.4804	17.5479	0.6319	14.7551	0.4619	17.1229	0.6146	14.6412	0.4556	17.0141	0.6102
MIST	800	16/255	15.8261	0.4955	17.5389	0.6104	14.2157	0.4156	16.2327	0.5575	13.8531	0.4001	15.8111	0.5393	13.776	0.3954	15.7018	0.5338
diffusionguard	800	16/255	15.6525	0.4855	16.5821	0.5574	14.4251	0.4325	15.2354	0.4981	14.1756	0.4232	14.9005	0.4824	14.0824	0.4182	14.8125	0.4783
ddd	800	16/255	15.0345	0.4869	16.5959	0.5805	13.354	0.4011	14.9779	0.5163	13.1464	0.384	14.5164	0.4943	13.1267	0.3806	14.4051	0.4886
Anti-Inpainting	800	16/255	14.3042	0.4613	14.7742	0.5058	13.1893	0.4145	13.4381	0.445	13.0784	0.4068	13.2167	0.4328	13.0593	0.4052	13.179	0.4304

TABLE VII
THE QUANTITATIVE RESULTS OF COMPARISON METHODS AND ANTI-INPAINTING UNDER RESIZE INTERPOLATION

Attack Method	iters	L_{∞} -eps	Resize Interpolation INTER_NEAREST				Resize Interpolation INTER_LINEAR				Resize Interpolation INTER_CUBIC			
			White-box Mask		Black-box Mask		White-box Mask		Black-box Mask		White-box Mask		Black-box Mask	
			PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓
photoguard	800	16/255	18.515	0.6635	19.4785	0.7088	18.7588	0.7136	19.8609	0.7551	17.2392	0.6472	18.8313	0.7065
advDM	800	16/255	18.902	0.6654	19.5316	0.7067	19.4159	0.733	20.1988	0.765	18.3577	0.6777	19.2818	0.723
MFA	800	16/255	19.2549	0.6767	19.9823	0.7229	19.8916	0.7549	21.0758	0.7917	18.7293	0.6992	20.0957	0.7519
MIST	800	16/255	18.9644	0.6656	19.3692	0.7015	19.9049	0.7495	20.7065	0.7803	18.5967	0.6866	19.8019	0.7405
diffusionguard	800	16/255	18.4713	0.6468	19.1041	0.6898	19.5782	0.7424	20.6111	0.7774	18.4414	0.6835	19.4714	0.729
ddd	800	16/255	17.9722	0.6313	18.5194	0.6701	18.4042	0.7048	19.3861	0.7411	17.1255	0.6344	18.4245	0.6943
Anti-Inpainting	800	16/255	17.247	0.6008	17.2927	0.6204	17.5147	0.6596	18.0567	0.6877	16.4458	0.603	17.0716	0.6372

dataset, surpassing all comparison methods. In our approach, we diversify the training conditions by incorporating initial latent state resampling and mask augmentation. The experimental results indicate that this strategy not only enhances the effectiveness of adversarial examples across various conditions but also mitigates the risk of overfitting to specific model weights. Note that our comprehensive cross-model transferability evaluation is conducted primarily on the diverse natural images of Inpaint-GuardBench. While similar transferability trends are expected on domain-specific datasets like CelebA-HQ, they are omitted here for brevity.

E. Robustness Study

To thoroughly evaluate the robustness of our proposed method, we conduct a series of comprehensive experiments. First, to investigate resilience against conventional image processing, we assess the performance of our approach and baseline methods under traditional lossy operations. Furthermore, we extend our evaluation to examine the robustness of our defense against recent, advanced perturbation purification models.

1) *Traditional Lossy Operations*: We simulate distinct distortion scenarios to rigorously test our approach. Specifically, we apply JPEG compression with quality factors of 90, 75,

and 50 to assess resilience against compression artifacts; we perform image resizing by downsampling images by 50% and subsequently upsampling them to their original dimensions using nearest-neighbor, linear, and cubic interpolation; and we implement 1-bit, 2-bit, 3-bit, and 4-bit quantization to mimic information loss in storage-constrained environments.

As summarized in Tables V, VI, and VII, our approach demonstrates superior resilience against the aforementioned traditional lossy operations, including JPEG compression, bit depth reduction, and image resizing. Furthermore, the qualitative results under JPEG compression scenarios corroborate our quantitative findings in Fig. 5. As visually demonstrated, our approach consistently maintains a robust protective effect across different compression quality factors. Even under varying degrees of compression, the images protected by Anti-Inpainting successfully induce noticeable visual distortions in the final manipulated outputs, thereby demonstrating the strong resilience of our adversarial perturbations.

Traditional proactive defenses typically overfit to the high-frequency features processed in the U-Net’s downsampling blocks, making them highly sensitive to operations that disrupt pixel-level details. In contrast, by perturbing features across the entire hierarchy—including downsampling, middle, and upsampling blocks—our approach ensures that the adversarial

TABLE VIII
THE QUANTITATIVE RESULTS OF COMPARISON METHODS AND ANTI-INPAINTING UNDER PERTURBATION PURIFICATION

Attack Method	iters	L_∞ -eps	DiffPure				GrIDPure			
			White-box Mask		Black-box Mask		White-box Mask		Black-box Mask	
			PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓	PSNR↓	SSIM↓
photoguard	800	16/255	18.3043	0.6592	17.9231	0.6504	16.1614	0.5911	16.1469	0.5923
advDM	800	16/255	18.3592	0.6634	17.9433	0.65	16.1873	0.5933	16.181	0.5929
MFA	800	16/255	18.3015	0.6590	17.9213	0.6511	16.0401	0.5875	16.0649	0.5898
MIST	800	16/255	18.292	0.6573	17.9472	0.6497	16.0167	0.5826	16.0353	0.5864
diffusionguard	800	16/255	18.3096	0.6603	17.855	0.6476	16.0139	0.5807	16.0218	0.5846
ddd	800	16/255	18.3576	0.6606	17.8938	0.6492	16.1712	0.5914	16.0997	0.5899
Anti-Inpainting	800	16/255	18.3594	0.6617	17.8534	0.6469	16.1425	0.5894	16.1074	0.59



Fig. 5. The qualitative results between comparison methods and Anti-Inpainting under JPEG Compression. The dashed regions highlight the primary subjects. While the manipulation model can easily recognize these subjects and synthesize matching backgrounds in the results of other methods, our approach prevents the model from effectively identifying the subjects, thereby generating incongruous backgrounds.

influence persists even when high-frequency information is removed. This comprehensive attack surface allows the protective perturbations to survive mid-to-high-level semantic transitions, ensuring consistent defensive efficacy [37].

2) *Diffusion-Based Reconstruction*: We further evaluate the defense against state-of-the-art perturbation purification frameworks, including DiffPure and GridPure, which pose a significant challenge to proactive protection.

As reported in Table VIII, the protective performance of all tested adversarial methods, including our own, experiences a significant decline when confronted with diffusion-based reconstruction operations (e.g., DiffPure and GridPure). Despite this drop in absolute efficacy, our approach still maintains comparable performance relative to other baseline defenses under these extreme conditions. Unlike simple pixel-wise transformations, these advanced purification techniques fundamentally re-generate the image content via a forward-and-backward diffusion process, which effectively “washes away” most adversarial signals. Ultimately, the substantial performance degradation observed across all existing methods against such advanced purification highlights a critical open challenge in proactive defense, which warrants further investigation in future studies.

The significant performance degradation observed against advanced diffusion-based purifications, such as DiffPure and GrIDPure, highlights a fundamental challenge in current proactive defenses. These purification frameworks are highly effective because their forward diffusion process inherently acts as a strong low-pass filter, systematically corrupting the carefully

crafted, high-frequency adversarial perturbations added at multiple levels. Consequently, the reverse denoising process projects the image back onto the clean data manifold, effectively washing out the fragile adversarial features before they can influence the target generation stages. To counter this in future work, proactive defenses must evolve beyond static perturbations. Potential directions include optimizing ‘purification-robust’ perturbations by targeting the low-frequency components of the image, which are more resilient to the forward diffusion process. Additionally, integrating the purification process itself into the adversarial optimization loop, could yield perturbations capable of surviving the diffusion-based denoising pipeline.

V. CONCLUSION

This paper introduces Anti-Inpainting, a novel proactive defense approach against malicious diffusion-based inpainting. Our approach integrates three key innovations to effectively protect images under unknown operational conditions. First, a multi-level deep feature extractor is utilized to enhance protective efficacy. Second, multi-scale semantic-preserving data augmentation is incorporated to improve the transferability of adversarial perturbations across diverse guidance conditions. Finally, a selection-based distribution deviation optimization strategy is developed to dynamically adjust the adversarial noise, mitigating ineffective updates. Extensive experiments demonstrate that Anti-Inpainting is a powerful proactive defense against malicious inpainting manipulations.

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